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B.Tech. Degree VI Semester Special Supplementary Examination November 2022

ME 19-205-0603 OPERATIONS MANAGEMENT (2019 Scheme)

Time: 3 Hours

Maximum Marks: 60

Course Outcome

On successful completion of the course, the students will be able to:

CO1: Understand different network techniques and forecasting methods.

CO2: Learn production planning and control as well as inventory control techniques.

CO3: Study the concepts of aggregate planning and scheduling.

CO4: Get the basic knowledge of plant location and lay out and material handling.

Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze,

L5 – Evaluate, L6 – Create

PO – Programme Outcome

PART A

(Answer *ALL* questions)

	(8 × 3 = 24)	Marks	BL	CO	PO
I. (a) Differentiate between PERT and CPM in network construction. List atleast one applications of each method.	3	3	L1	1	1
(b) What is crashing of network in project network? Explain with a suitable example.	3	3	L3	1	1
(c) What are the objectives of production planning and control?	3	3	L2	2	1
(d) Explain standardization and simplification in production planning.	3	3	L3	2	3
(e) What is the purpose of Gantt chart?	3	3	L1	3	1
(f) What is Aggregate planning?	3	3	L2	3	
(g) What are the different types of maintenance methods used in production industry?	3	3	L2	4	1
(h) What are the factors to be considered while selecting a plant location?	3	3	L3	4	2

PART B

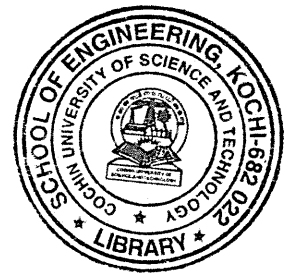
(4 × 12 = 48)

- II. A project consists of 4 activities. Their logical relationship and time taken is given along with crash time and cost details. If the indirect cost is ₹2000/- per week, find the optimal duration and optimal cost.

Activity	Predecessor	Normal Time		Crash Time	
		Time in days	Cost in ₹	Time in days	Cost in ₹
A	-	4	4000	2	12000
B	A	5	3000	2	7500
C	A	7	3600	5	6000
D	B	4	5000	2	10000
Total			15600		35500

OR

(P.T.O.)



BTS-VI(S.S)-11-22-1119

- III. Draw the network for the given project and find the probable project completion time.

Marks 12 BL L4 CO 1 PO 11

Activities		Time in Days		
i	j	to	tm	tp
1	2	5	12	17
1	3	8	10	13
1	4	9	11	12
2	3	5	8	9
2	5	9	11	13
4	6	14	18	22
3	7	21	25	30
6	7	8	13	17
6	8	14	17	21
7	8	6	9	12

- IV. (i) Explain the term product life cycle with the help of a diagram. 6 L3 2 1
(ii) What are the factors affecting product design? Explain. 6

OR

- V. (i) What is just in time and lean management. 6 L4 2 2
(ii) How a material requisition helps in procurement policy. 6
Explain the procedure adopted by a manufacturing industry for implementing procurement policy based on material requisition.

- VI. (i) What is sequencing? How it is applied in m machines - n jobs problems. 6 L4 3 2
(ii) Find the sequence of operation that minimise the total elapse time required to complete the job on machines M1 and M2 in the order M1M2. The processing time to complete the job in hours is given below. Also find the total elapsed time and the idle time for each machine 6

Job	A	B	C	D	E	F	G
M1	5	2	9	3	10	7	6
M2	2	6	7	8	4	4	3

OR

- VII. What is critical ratio in scheduling of jobs? Explain with a simple example of your choice. 12 L2 3 1

- VIII. (i) What is depreciation? How it affect the replacement policy? 12 L4 4 2
(ii) A firm is thinking of replacing a particular machine whose cost price is ₹25,200. The scrap value of the machine is ₹500/-. The maintenance costs are found to be as follows:

Year	1	2	3	4	5	6	7	8
Maintenance cost in ₹	420	700	1000	1300	1900	2500	3200	4000

Determine when the firm should get the machine replaced.

OR

- IX. Explain the different types of plant layout used in industry. Which layout is commonly used in coal industry? Why? 12 L4 4 2

Bloom's Taxonomy Levels

L1 = 5%, L2 = 17.5%, L3 = 27.5%, L4 = 50%.

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B.Tech. Degree VI Semester Special Supplementary Examination November 2022

ME 19-205-0603 OPERATIONS MANAGEMENT (2019 Scheme)

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Maximum Marks: 60

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PO – Programme Outcome

PART A

(Answer *ALL* questions)

	(8 × 3 = 24)	Marks	BL	CO	PO
I. (a)	Differentiate between PERT and CPM in network construction. List atleast one applications of each method.	3	L1	1	1
(b)	What is crashing of network in project network? Explain with a suitable example.	3	L3	1	1
(c)	What are the objectives of production planning and control?	3	L2	2	1
(d)	Explain standardization and simplification in production planning.	3	L3	2	3
(e)	What is the purpose of Gantt chart?	3	L1	3	1
(f)	What is Aggregate planning?	3	L2	3	
(g)	What are the different types of maintenance methods used in production industry?	3	L2	4	1
(h)	What are the factors to be considered while selecting a plant location?	3	L3	4	2

PART B

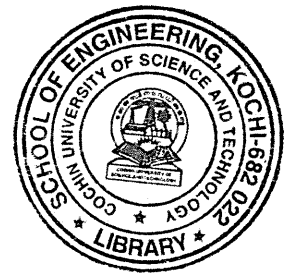
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(P.T.O.)



BTS-VI(S.S)-11-22-1119

- III. Draw the network for the given project and find the probable project completion time.

Marks 12 BL L4 CO 1 PO 11

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1	4	9	11	12
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3	7	21	25	30
6	7	8	13	17
6	8	14	17	21
7	8	6	9	12

- IV. (i) Explain the term product life cycle with the help of a diagram. 6 L3 2 1
(ii) What are the factors affecting product design? Explain. 6

OR

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Job	A	B	C	D	E	F	G
M1	5	2	9	3	10	7	6
M2	2	6	7	8	4	4	3

OR

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(ii) A firm is thinking of replacing a particular machine whose cost price is ₹25,200. The scrap value of the machine is ₹500/-. The maintenance costs are found to be as follows:

Year	1	2	3	4	5	6	7	8
Maintenance cost in ₹	420	700	1000	1300	1900	2500	3200	4000

Determine when the firm should get the machine replaced.

OR

- IX. Explain the different types of plant layout used in industry. Which layout is commonly used in coal industry? Why? 12 L4 4 2

Bloom's Taxonomy Levels

L1 = 5%, L2 = 17.5%, L3 = 27.5%, L4 = 50%.

B.Tech. Degree VI Semester Examination April 2018

ME 15-1603 OPERATIONS MANAGEMENT (2015 Scheme)

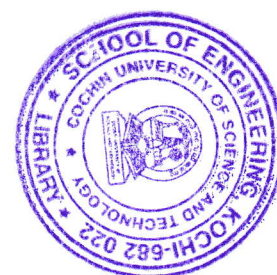
Time: 3 Hours

Maximum Marks: 60

PART A (Answer *ALL* questions)

(10 × 2 = 20)

- I. (a) Define operations management and list its functions.
 (b) Distinguish between PERT and CPM.
 (c) Differentiate between prediction and forecasting.
 (d) Define loading, scheduling and expediting in PPC.
 (e) Write notes on A.B.C and VED methods of inventory control.
 (f) Explain bin cards with its layout.
 (g) Explain Gantt charts with a diagram.
 (h) What are the different priority rules in scheduling?
 (i) List the factors to be considered for locating a factory.
 (j) Define the term depreciation and list the methods of calculating depreciation.



PART B

(4 × 10 = 40)

- II. Draw the network for the following data, find the critical path and project duration. Also find the total float, for the activities. (10)

Activity	A	B	C	D	E	F	G	H	I	J	K	L	M	N
Predecessor	-	A	A	C	C	B, G	D	D	D	F, H	I	E, K	J	L
Time-wks	10	6	2	12	9	4	8	10	5	5	0	7	10	8

OR

- III. (a) Define the term crashing of network, explain with a simple example of your choice. (5)
 (b) Explain the exponential smoothing method of forecasting with an example. (5)
- IV. (a) Define the terms routing and scheduling. Discuss them for (i) continuous (ii) assembly and (iii) job order industry. (5)
 (b) Discuss the procedure you would follow for planning and control of production for a steel furniture industry when orders are received from customers. (5)

OR

- V. (a) Derive the expression for EOQ for basic inventory model and list the assumption used. (5)
 (b) In a manufacturing firm the annual requirement is 24000 units, the supply is instantaneous and shortages are allowed. Each ordering costs ₹350, the inventory holding cost is ₹0.10 per unit per month and the cost of shortage is ₹0.20 per unit per month. Determine the EOQ and the total cost of inventory for this quantity. (5)

(P.T.O.)

- VI. Six jobs are to be scheduled in three machines. The processing time is as follows. Find the optimal schedule so that total elapsed time is minimized. Draw the Gantt chart for the three machines. (10)

Job	J1	J2	J3	J4	J5	J6
M/c1	10	3	5	4	2	1
M/c2	2	4	6	3	1	2
M/c3	8	6	7	9	7	7

OR

- VII. Five jobs arrive at processing station one after another. The processing time and due date are as follows. Use sequencing rules to find (i) total flow time (ii) mean flow time and lateness of job and tell which method is best among. (10)

Job in the sequence of arrival	J1	J2	J3	J4	J5
Processing time days	3	4	2	6	1
Due date (days hence)	5	6	7	9	2

- VIII. (a) Explain the selection factors to be considered for locating a plant. (5)
 (b) What are the most common approaches for plant layout? Compare them. (5)

OR

- IX. (a) What factors influence the selection of material handling equipment? State the application of the following. (5)
 (i) Fork lift truck (ii) Belt conveyor (iii) Jib crane (iv) Roller conveyor.
 (b) The cost of an equipment is ₹7200 and the scrap value is ₹200, the maintenance cost are as follows. When the equipment should be replaced? (5)

Year	1	2	3	4	5	6	7	8
Annual maintenance cost	150	300	450	650	950	1300	1850	2500



B.Tech. Degree VI Semester Regular Examination April 2022

ME 19-205-0603 OPERATIONS MANAGEMENT (2019 Scheme)

Time: 3 Hours

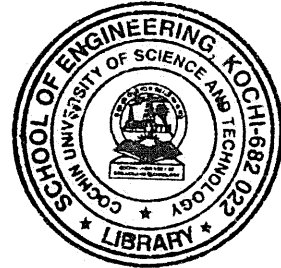
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Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze,
 L5 – Evaluate, L6 – Create
 PO – Programme Outcome



PART A (Answer *ALL* questions)

- | | | Marks | BL | CO | PO |
|-----|--|-------|----|----|----|
| I. | (8 × 3 = 24) | | | | |
| (a) | Sketch a dummy activity in network and mention its importance with respect to the succeeding activity. | 3 | L1 | 1 | 2 |
| (b) | What is the importance of demand forecasting in operations management? | 3 | L2 | 1 | 1 |
| (c) | Discuss the relationship between simplification, standardisation and interchangeability. | 3 | L4 | 2 | 2 |
| (d) | What is a bin card? Sketch and explain its use. | 3 | L2 | 2 | 2 |
| (e) | What is the role of aggregate planning in manufacturing operations? | 3 | L2 | 3 | 2 |
| (f) | What are the normal rules adopted for priority sequencing? | 3 | L1 | 3 | 1 |
| (g) | List out the significance of locating manufacturing unit in rural areas. | 3 | L1 | 4 | 2 |
| (h) | Differentiate between maintenance and replacement of machineries. | 3 | L4 | 4 | 2 |

PART B

(4 × 12 = 48)

- | | | | | | |
|-----------|--|---|----|---|---|
| II. | (a) Discuss the relationship of production function with other business activities of an Industry. | 6 | L3 | 1 | 2 |
| | (b) How forecasting methods are classified? Give examples for each. | 6 | L1 | 1 | 1 |
| OR | | | | | |
| III. | (a) Differentiate between any two project network analysis methods. | 3 | L2 | 1 | 2 |
| | (b) Construct the network diagram for the project given below. Also determine the critical path, expected project duration and probability of completing the project in 55 days. | 9 | L5 | 1 | 1 |

Activity	Immediate predecessor	Time (Days)		
		to	tm	tp
A	-	4	6	8
B	A	5	7	15
C	A	4	8	12
D	B	15	20	25
E	B	10	18	26
F	C	8	9	16
G	E	4	8	12
H	D,F	1	2	3
I	G,H	6	7	8

BTS-VI(R)-04-22-(SC)-0648

IV.	(a)	Explain the steps involved in product design with the help of a flow chart.	6	L3	2	1
	(b)	What are the objectives of production planning and control? Discuss.	6	L2	2	2
OR						
V.	(a)	How industrial inventories are classified? Briefly discuss about VED and FSN analysis of Inventory management.	6	L2	2	1
	(b)	Monthly consumption of an item is 300 units. The price per unit is Rs.10. Inventory carrying cost is 18 per cent and ordering cost is 36 per order. Lead time of 1 month stock. Determine Re-order quantity, Re-order level, Minimum and maximum stock level, Average inventory.	6	L5	2	2
VI.		With a suitable example, explain the graphical and reaction rate methods of aggregate planning.	12	L4	3	3
OR						
VII.	(a)	What are the factors affecting scheduling decisions in manufacturing?	4	L1	3	1
	(b)	Discuss the procedure and conditions to be satisfied while performing sequencing of n Jobs and 3 machine problem with an example of your choice.	8	L4	3	2
VIII.	(a)	Explain the steps involved in plant layout design process.	6	L3	4	1
	(b)	List and explain the factors influencing plant location decisions.	6	L2	4	2
OR						
IX.	(a)	How material handling equipment's are classified? Explain with examples.	6	L1	4	2
	(b)	List and briefly explain the features of different maintenance methods (minimum 4 methods).	6	L2	4	2

Blooms's Taxonomy Levels

L1 = 20.83%, L2 = 30.00%, L3 = 15.00%, L4 = 21.66%, L5 = 12.50%

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B.Tech. Degree VI Semester Regular Examination April 2022

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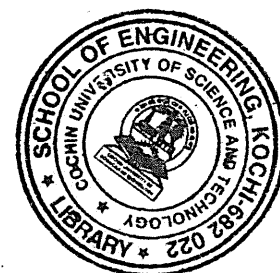
CO3: Study the concepts of aggregate planning and scheduling

CO4: Get the basic knowledge of plant location and lay out and material handling

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PO – Programme Outcome



PART A

(Answer *ALL* questions)

I.	(8 × 3 = 24)	Marks	BL	CO	PO
(a) What is critical Path in network analysis?		3	L1	1	1
(b) Differentiate between qualitative Vs quantitative forecasting.		3	L2	1	1
(c) Analyse how product life cycle is important in industry.		3	L4	2	1
(d) How Just in Time (JIT) manufacturing strategy helps in inventory control.		3	L3	2	2
(e) What is the need for aggregate planning?		3	L1	3	1
(f) Discuss critical ratio method of loading and scheduling of jobs to machines.		3	L1	3	2
(g) Identify factors that decides whether to go for maintenance of equipments or replacement.		3	L5	4	2
(h) Explain the concept of depreciation.		3	L1	4	1

PART B

(4 × 12 = 48)

II. (a) Explain production process as a system and its various components briefly.	4	L2	1	1
(b) Draw network of the following project and identify the critical path.	8	L3	1	2

Activity	A	B	C	D	E	F	G	H	I	J	K	L	M
Predecessors	-	A	B	A	D	E	-	G	J, H	-	A	C, K	I, L
Duration(Days)	6	4	7	2	4	10	2	10	6	13	9	3	5

Also, find out the project duration.

OR

(P.T.O)

BTS-VI(R)-04-22-0296

- III. (a) Generate the forecasted sale values of following time series data using a simple moving average (SMA) and with the weighted moving average (WMA) with averaging interval (m) = 3 for both cases. For WMA, the more recent observations are weighted more than older observations using the weights $w_1 = 0.6$, $w_2 = 0.3$ and $w_3 = 0.1$. 8 L6 1 2

t	Time Bucket (Month)	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Y	Actual Sales in '000)	3	5	9	20	12	17	22	23	51	41	56	75	60	75	88

Compute the Mean Absolute Error (MAE) and Mean Square Error (MSE) of SMA and WMA. Comment on forecast obtained.

- (b) Discuss briefly: (i) slack (ii) multi-time estimate in PERT. 4 L1 1 1
- IV. (a) Distinguish briefly simplification, standardization, specialisation and inter-changeability. 6 L1 2 1
- (b) Purchasing department of a company estimates carrying cost as 15% and ordering cost at \$90 per order. The estimated annual requirement is 78,000 units at a price of \$4 per unit. 6 L3 2 2
- (i) What is the most economical number of units to order?
- (ii) Determine number of orders to be placed in a year.
- (iii) Find about how often will an order need to be placed.

OR

- V. (a) Discuss objectives and functions of PPC (Production Planning and Control). 6 L2 2 1
- (b) Explain EOQ model of inventory control. Derive the equation for EOQ. 6 L3 2 1
- VI. (a) Densepack, a disk drive producer has 300 workers employed at the moment. The ending inventory in December 500 units. The firm would like to have 600 units at the end of June. Inventory requirements and on-hand inventories are dealt with by adding/subtracting them to the expected demand. The expected demand based on forecast is as follows 8 L4 3 2

Demand and production data

Month ↓ Description →	Predicted demand	Net predicted demand	Net Cumulative demand	Number of Working days
January	1280	780	780	20
February	640	640	1420	24
March	900	900	2320	18
April	1200	1200	3520	26
May	2000	2000	5520	22
June	1400	2000	7520	15

Cost data

C_H : cost of hiring one worker = \$ 500; C_F : cost of firing one worker = \$ 1000;

C_I : cost of holding one unit of inventory one month = \$ 80; Salary to each worker: \$ 2500 per month. K ; Number of aggregate units produced by one worker in one day: It was observed that 76 workers produced 245 disk drives in 22; $K = 245/(76 \times 22) = 0.14653$

Strategy 1: Chase the demand by changing work force levels

Strategy 2: Keep the work force level constant throughout the planning horizon
Compare the total cost associated with above two Aggregate Planning Strategies. Draw graphs between months (X-axis) and Cumulative demand and production (Y-axis) for both plans and commend on the feasibility of the plans.

- (b) Explain Gantt Charts with sketches. 4 L2 3 1

OR

(Continued to 3)

BTS-VI(R)-04-22-0296

VII. (a)

JOB	A	B	C	D	E	F
Processing Time (t)	3	12	9	11	6	4
Due Date (D)	7	16	4	17	15	18

8 L3 3 2

The above independent jobs are sequenced into workstation with multiple job processing capacity from A to F. If jobs are sequenced under Short Processing Time (SPT), Due Dates (DD) and Critical ratio (CR) priority rules into the workstation, find the following for each rule.

1. Job Sequence
2. Total Tardiness
3. Total flow time
4. Average tardiness
5. Average flow time

(b) Explain job sequence using Johnson's rule?

4 L1 3 1

VIII. (a) Explain the following principles of Plant Layout briefly.

8 L2 4 1

- (i) Principle of Minimum Movement
- (ii) Principle of Space Utilization
- (iii) Principle of Flexibility
- (iv) Principle of Interdependence
- (v) Principle of Overall Integration
- (vi) Principle of Safety
- (vii) Principle of Smooth Flow
- (viii) Principle of Economy

(b) What are the factors to be considered for equipment replacement?

4 L2 4 1

OR

IX. (a) What is Systematic Plant Layout (SLP)? Explain the fundamental concepts for SLP

6 L1 4 1

(b) Explain briefly factors influencing plant Locations.

6 L2 4 1

Bloom's Taxonomy Levels

L1: 33.3%, L2: 29.2%, L3: 20.8%, L4: 8.3%, L5: 4.2%, L6: 4.2%
